SINGAPORE SCHOOL KINDERLAND

Computer Science — Official Teacher Answer Key & Mark Scheme

Topic 1.2.3 & 1.2.4: Data Compression • Total Marks: 25

Question 1: Multiple Choice Questions (5 Marks)

1. **ii.** To reduce transmission time and save bandwidth [1 mark]
2. **ii.** Redundant data is permanently eliminated (original cannot be reconstructed) [1 mark]
3. **i.** Removing sounds outside the human hearing range and using perceptual music shaping [1 mark]
4. **ii.** Less sensitive to colour variations than brightness [1 mark]
5. **iii.** A software program executable (`game.exe`) [1 mark]

Question 2: Classification Table (5 Marks)

No.	Scenario / File Type	Lossy (✓)	Lossless (✓)	Exam Explanation
i.	Streaming an HD movie over Netflix (`.mp4`)	✓		High compression needed; human eye does not detect minor losses.
ii.	A company financial accounting spreadsheet (`.xlsx`)		✓	Data loss would corrupt formulas and financial figures.
iii.	Uploading a high-resolution photo to social media (`.jpg`)	✓		JPEG permanently discards inaudible/subtle color details.
iv.	Downloading an operating system setup file (`setup.exe`)		✓	Executable program files must have 100% exact binary accuracy.
v.	Listening to music tracks on a smartphone (`.mp3`)	✓		90% reduction via perceptual music shaping.

Question 3: Run-Length Encoding (RLE) Text Calculations (6 Marks)

a) String: `X X X X X X  Y Y Y Y  Z Z Z Z Z Z Z Z`

- **i.** 6 ('X') + 4 ('Y') + 8 ('Z') = **18 characters = 18 Bytes**. [1 mark]
- **ii.** `6X 4Y 8Z` (or in ASCII decimal: `06 88  04 89  08 90`). [2 marks]
- **iii.** 3 pairs × 2 bytes = **6 Bytes**. [1 mark]
  Percentage Reduction = `((18 - 6) / 18) × 100% = 66.67%` size reduction. [1 mark]

b) Why RLE is ineffective on non-repeating string:
There are no repeated adjacent characters (each run length is only 1). Encoding each character with a count pair would produce 16 bytes (`1A 1B 1C...`) for an 8-byte string, **doubling the file size** rather than compressing it. [1 mark]

Question 4: Image Compression & RLE (5 Marks)

- **i.** Row 1 (8 bytes) + Row 2 (8 bytes) = **16 Bytes** uncompressed. [1 mark]
- **ii.** Row 1 RLE: `8B` | Row 2 RLE: `2B 6W`. [2 marks]

- **iii.** Total RLE pairs = 1 pair (Row 1) + 2 pairs (Row 2) = 3 pairs. Total compressed size = 3 × 2 = **6 Bytes**. Saving: Reduced from 16 Bytes to 6 Bytes (saving of 10 Bytes or **62.5%**). [2 marks]

Question 5: Short Answer Questions & BCD (4 Marks)

- **a)** Removing inaudible frequencies discards sound waves outside the 20 Hz – 20 kHz human hearing range [1 mark]. Auditory masking discards quieter sounds that play at the exact same time as louder sounds because the human ear cannot perceive the quieter sound [1 mark].
- **b) Representing 359.2 in BCD:**
  3 = `0011`, 5 = `0101`, 9 = `1001`, 2 = `0010` [1 mark].
  Final String: `0011 0101 1001 . 0010` [1 mark].